

G02 / ASSESSMENT

G02 GATE A

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

YOUR STANDALONE GATE BOOK

Independent SQL + PostgreSQL checkpoint

Assessment rule: Save the current attempt before opening its review. Fresh Gate B stays closed until explicitly assigned after repair.

When to attempt

Complete M02, M03 and their changed retries, including the integrated challenge. G02 measures whether you can reason from a new delivery, implement the intended result and explain the evidence. It is not passed by watching all videos or reading this book. Mentor automation is paused; use the supplied study record and save files manually.

Use only `course.g02_customers`, `course.g02_orders` and `course.g02_items` for the main assignment. They were supplied by `labs/SQL/01_load_assessments.sql`. These are new fictional inputs, not a tutorial dataset. Inspect their definitions and values as part of the task. Keep the protected review closed until your actual first attempt, output and uncertainty record are saved.

Contract and permitted help

The customer table has one row per `customer_id` with a known region. The order table has one row per `order_id` and may lack customer identity. The item table has one row per (`order_id`, `line_no`). Gross item amount is quantity multiplied by `unit_price`. Order gross is the sum of an order's items plus its `shipping_fee` exactly once. No tax or discount is applied. An order without item rows is intentionally empty in this complete fixture; its item total is zero and any stored shipping fee remains payable. NULL payment is unknown, not unpaid.

You may inspect source data, field definitions and error messages. For an independent gate attempt, do not open solved lesson queries or ask AI to write the answer. If you need conceptual help, record it honestly, save the attempt and use the repair process. An assisted learning attempt is useful but is not independent evidence. There is no required speed target; note actual elapsed time if it helps you plan later practice.

Assignment: save queries in numbered sections

1. Inspect and record the grain, keys, input row counts, item gross total and original shipping-fee total. Identify known-paid, known-unpaid and unknown-payment orders separately. Query orders on December 1 using an inclusive beginning and an exclusive next-day boundary.
2. Build a reusable query with named CTE stages returning one row per order. Include `order_id`, `customer_id`, `ordered_on`, region group, payment group, `item_gross`, `shipping_fee` and `order_gross`. Preserve every order. Demonstrate unique output IDs and an independent reconciliation to the two original amount sources.
3. Produce a region summary containing `order_count`, `item_gross`, `shipping_fees`, `order_gross` and average gross per order. Return groups whose gross is at least 100 in a second query. Explain the scope difference between the full and thresholded summaries.

4. Diagnose a raw orders-to-items join. Report joined row count, represented distinct order count and repeated shipping-fee sum. Explain precisely where repetition and missing order coverage occur. Explain why SUM(DISTINCT shipping_fee) over the order source also fails as a fee-total rule.
5. Return the highest-gross order per known customer with ROW_NUMBER, resolving amount ties by ascending order_id. In a separate query, return latest orders per known customer by date descending and ID descending. Explain why "highest" and "latest" use different orderings and can choose different rows.
6. Build two one-column lists of customer IDs: customers with a known-paid order and customers with an unknown-payment order. Combine them with UNION ALL and then UNION, describing the difference in occurrences versus unique selected IDs. Find customers with no orders using NOT EXISTS.
7. In a disposable temporary table, define a non-null positive quantity and a primary key. Demonstrate one rejected duplicate and one rejected negative quantity. In a separate explicit transaction, update a valid row, inspect it, ROLLBACK and verify restoration. Use EXPLAIN on one read-only query and explain estimated versus actual execution evidence. Identify your server version, connected database and role without revealing a password.
8. Save a short explanation of output grain, unknown handling, empty-order handling, tie rules and the denominator of your overall average. Describe one realistic changed input that would break a careless implementation. This explanation is assessed along with the queries, not after them as optional decoration.

Evidence to keep

Create G02_A_attempt.sql and G02_A_evidence.txt in your own work folder. Keep actual outputs or exported results, error messages for deliberate constraint failures, the source counts and totals, help used and unresolved issues. Save a separate G02_A_repair.sql if you revise after review. A result number typed from the answer book is not execution evidence.

Scoring every dimension separately

Use levels 0 to 4. Level 0 means no usable attempt; 1 means major misunderstanding or a copied solution; 2 means partly correct or dependent on guidance; 3 means independently correct with the required checks and explanation; 4 means level 3 plus a successful relevant changed-input or boundary test and a clear explanation of why it matters. Every dimension must reach at least 3, with no critical failure. Do not average scores to hide a weak dimension.

Protected dimension	Evidence required for level 3
Foundation SQL semantics	Correct filters, dates, NULL states and meaning of counts in tasks 1 and 8

Transformation + grouped reasoning

Correct expressions, CASE, grouped totals and HAVING scope in tasks 2 and 3

Joins, cardinality + set logic	Explained join multiplication, retained coverage, UNION difference and absence query in tasks 4 and 6
CTEs + window/ranking	Checkable named stages and correct deterministic highest/latest selection in tasks 2 and 5
Multi-grain reconciliation	Fees once per order, independent source reconciliation, correct averages and empty/unknown handling
PostgreSQL database thinking	Correct identity/range/null constraints, actual rollback evidence, connection awareness and honest plan interpretation
Evidence + explanation discipline	Saved original attempt, observed outputs, declared help, specific reasoning and separate repairs

Critical failures include multiplied financial amounts, wrong final grain, presenting copied or assisted answers as independent, or accepting unvalidated totals. Losing required orders or claiming an unknown payment is known also blocks acceptance until corrected. A neat PDF or an average score above three cannot override these failures.

Review and repair rule

Only after saving the first attempt, open the G02-A explained review. Compare reasoning and actual evidence, then identify each weak dimension. Use the changed micro-task on `course.g02_repair_*` for that dimension, with the relevant query constructed again from a blank editor. A minor repair does not require restarting every strong dimension, but each failed dimension needs fresh independent evidence. If you need a complete new gate after extensive solution exposure, full variants B and C remain reserved for deliberate later assignment; they are not supplied or falsely marked delivered in this batch.

Passing this gate means readiness to enter the next standalone module's prerequisites. It is not a hiring guarantee or an external certification. If you can reproduce numbers only with the review open, continue targeted practice before recording level 3.